

$$1 \quad (A+B)(C+B)(C+A)$$

$$= (AC + AB + BC + BB)(C+A)$$

$$= ACC + ABC + BCC + BBC + AAC + AAB + ABC + ABB$$

$$= AC + ABC + BC + AB$$

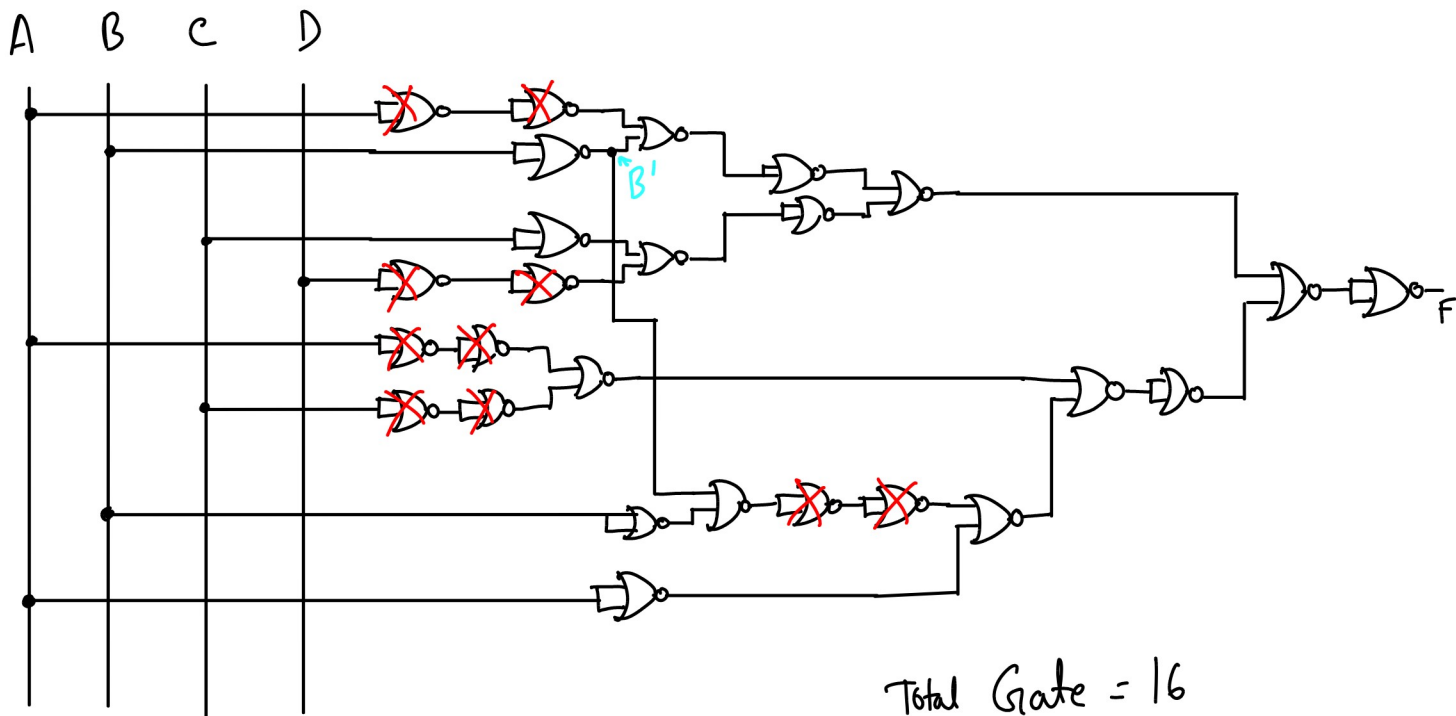
$$= AC + BC(A+1) + AB$$

$$= AC + BC + AB$$

$$= A(B+C) + BC$$

3.

$$F(A,B,C,D) = (A'BCD' + A'C' + A(B'+D'))$$



$$5. F(A, c, B) = A'c + Bc'$$

$$\begin{aligned} \text{SOP,} &= A'c (BB') + Bc' (A \cdot A') \\ &= A'cB + A'cB' + Ac'B + A'c'B \\ &= \sum (011, 010, 101, 001) \\ &= \sum (3, 2, 5, 1) \\ &= \sum (1, 2, 3, 5) \end{aligned}$$

$$\text{POS, } F(A, c, B) = A'c + Bc'$$

$$\begin{aligned} &= (A'c + B) (A'c + c') \\ &= (B + A') (B + c) (A' + c') (c + c') \\ &= (A' + B + cc') (B + c + AA') (A' + c' + BB') \cdot 1 \\ &= (A' + B + c) (A' + B + c') (A + B + c) \cancel{(A' + B + c)} \\ &\quad \cancel{(A' + B + c')} (A' + B' + c') \\ &= (A' + c + B) (A' + c' + B) (A + c + B) (A' + c' + B') \\ &= \pi (100, 110, 000, 111) \\ &= \pi (4, 6, 0, 7) \\ &= \pi (0, 4, 6, 7) \end{aligned}$$

$$6. F(A, B, C, D) = A + B'C' + D$$

$$\text{SOP} = A + B'C' + D$$

$$= A(B+B')(C+C')(D+D') + B'C'(A+A')(D+D') + D(A+A')(B+B')(C+C')$$

$$= (AB+AB')(CD+CD'+C'D+C'D') + (AB'C'+A'B'C')(D+D') + (AD+AD')(BC+BC'+B'C+B'C')$$

$$= ABCD + ABCD' + ABC'D + ABC'D' + AB'CD + AB'CD' + AB'C'D + AB'C'D' +$$

$$AB'C'D + AB'C'D' + A'B'CD + A'B'CD' + A'B'C'D + A'B'C'D' + A'B'C'D + A'B'C'D'$$

$$= \sum (1111, 1110, 1101, 1100, 1011, 1010, 1001, 1000, 1001, 1000, 0001, 0000, 1111, 1011, 1101, 1001, 0111, 0011, 0101, 0001)$$

$$= \sum (15, 14, 13, 12, 11, 10, 9, 8, \cancel{7}, \cancel{6}, 1, 0, \cancel{15}, \cancel{14}, \cancel{13}, \cancel{9}, 7, 3, 5, \cancel{1})$$

$$= \sum (0, 1, 3, 5, 7, 8, 9, 10, 11, 12, 13, 14, 15)$$

$$\text{POS} = A + B'C' + D$$

$$= A + D + B'C'$$

$$= (A + D + B')(A + D + C')$$

$$= (A + D + B' + C)(A + D + C' + BB')$$

$$= (A + B' + C + D)(A + B' + C' + D)(A + B + C' + D)(\cancel{A + B' + C' + D})$$

$$= \pi(0100, 0110, 0010)$$

$$= \pi(4, 6, 2)$$

$$= \pi(2, 4, 6)$$